

# Danilov Sergey Dmitrievich

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> ML/LLMOps Engineer who owns the full lifecycle of production ML and LLM serving — model deployment, distributed inference, GPU orchestration, vLLM and VeRL, batching and scheduling, MoE routing, and predictive reliability of cluster hardware (memory, network) — turning models into reliable, high-availability services with measurable SLO.

## CAREER

### Huawei — LLM Inference R&D Engineer · *September 2024 – present*

- Disaggregated Prefill/Decode simulator — C++ discrete-event simulation of a prefill/decode serving system with KV-cache-aware load balancing and Markov request traffic; selects TP/DP/SP/EP topology and scheduling policy to hold 99.99% SLO, and steers vLLM recovery for RL rollouts.
- Time-series clustering of optical-power telemetry flags degrading links early; a minimal-parameter model that raised cluster availability by +10%.
- NUMA-aware stress testing reached 95% of theoretical bandwidth; Weibull MTTR modeling + XGBoost failure forecasting cut repair costs by 2×. doc
- Combinatorial EPLB MoE expert-placement algorithms that co-optimize expert-to-GPU mapping with all-to-all communication to minimize network congestion and tail latency.
- Deep knowledge across the serving stack: continuous batching and preemptive scheduling, fused communication operators, FP8/INT8 quantization, speculative decoding, and optimization trade-offs.
- LLMOps for an LLM quantization project: owned the full lifecycle from research notebook → quantized model → deployed service and benchmarks, speeding up the research cycle by 3×.

### ML Engineer — ITMO University · *November 2023 – September 2024*

- Sber & ITMO: CTGAN user emulator; improved BiVAE recommendation MAP@5 by 3%.
- Almazov & ITMO: Shapley-valued weighted XGBoost for obesity treatment; 71% accuracy for 3-month sibutramine therapy response; 2 Q2 papers. J. Pers. Med. J. Clin. Med.
- Industrial AI assistant: improved PDF NER with prompt engineering/self-consistency; filtering removed 80% repetition.

### Testing Engineer — Harman · *September 2021 – July 2022*

- Sped up testing for complex GPS automotive systems with Robot Framework

### Release Engineer — Intel · *July 2019 – August 2021*

- Supported Intel OneAPI HPC release infrastructure: Python/Bash automation and C++ installer.

## TECH STACK

- **AI infrastructure:** Harness, Agents workflow, RAG, CAG, vLLM, VeRL, PyTorch, Kubernetes
- **Data Analysis & Research:** Python, Pandas, Scikit-learn, AutoML, PatentScope, Elsevier Scopus

## EDUCATION

**ITMO University, 2024 — Master's degree in Data Analysis**

– Thesis: Shapley-value weighted filtering for small-sample ML; +7% accuracy on medical data.

**Higher School of Economics, 2022 — Bachelor's degree in Software Engineering**

– Thesis: accessible assistant for visually impaired users using Yandex Cloud OCR.

## PUBLICATIONS

- Google Scholar · h-index 2
- Danilov et al., J. Pers. Med. 2024, 14, 811 — sibutramine therapy prediction.
- Markina et al., J. Clin. Med. 2024, 13, 4151 — brown adipose tissue and obesity treatment.

## COMPETITIONS

**ScienceRAG — Research assistant agent based on GigaChat**

– ScienceRAG: improved ArXiv RAG retrieval on GigaChat YouTube

**X5 AI Tech Hackathon — Hallucination detection algorithm**

– X5 AI Tech: ruBERT hallucination detector — 88% accuracy, <0.45 s inference; top-10.

## LANGUAGES

- Russian, English (B1)